

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	16-April-2026	Section	6A , 6B
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Consider an IP address block where one of its address is 172.31.200.130/19. Please provide the following: **[1 * 5 = 5 Marks] [CLO 3]**

First Address of block: _____

Total number of addresses: _____

First Useable IP: _____

Last useable IP: _____

Broadcast Address: _____

Q2. Host A sends a single IPv4 datagram to Host B over a network path that contains a single intermediate router (R1). The original datagram exiting Host A has the following values in its IPv4 header: **[10 Marks] [CLO 3]**

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hex)	DF Flag	MF Flag	Fragment Offset
1500	20	0xABCD	0	0	0

The Maximum Transmission Unit (MTU) on the link connecting Host A to Router R1 is 1500 Bytes. The MTU on the link connecting Router R1 to destination Host B is 600 Bytes.

Draw a 6-column table (identical to the one above) and fill in the corresponding header details for *every* fragment created and forwarded by Router R1.

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	16-April-2026	Section	6A , 6B
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Consider an IP address block where one of its address is 192.168.45.77/21. Please provide the following: **[1 * 5 = 5 Marks] [CLO 3]**

First Address of block: _____

Total number of addresses: _____

First Useable IP: _____

Last useable IP: _____

Broadcast Address: _____

Q2. A home router connects a private network (192.168.1.0/24) to the ISP. The router's WAN (Public) IP is 203.0.113.5. Two internal devices establish connections simultaneously:
[6+4 = 10 Marks] [CLO 3]

- **Host A (192.168.1.10)** accesses a web server at 128.119.245.12:80 using source port 3345.
- **Host B (192.168.1.11)** accesses the same web server at 128.119.245.12:80 using source port 3345 (coincidentally the same source port).

Tasks:

1. Construct the **NAT Translation Table** that would exist inside the router to support this scenario. (Create columns for WAN Side and LAN Side).
2. Explain precisely how the router distinguishes the returning packets from the web server, ensuring Host A's traffic doesn't go to Host B.